

Devin Fromond

Lab Report

ECE 2031 L04

23 September 2025

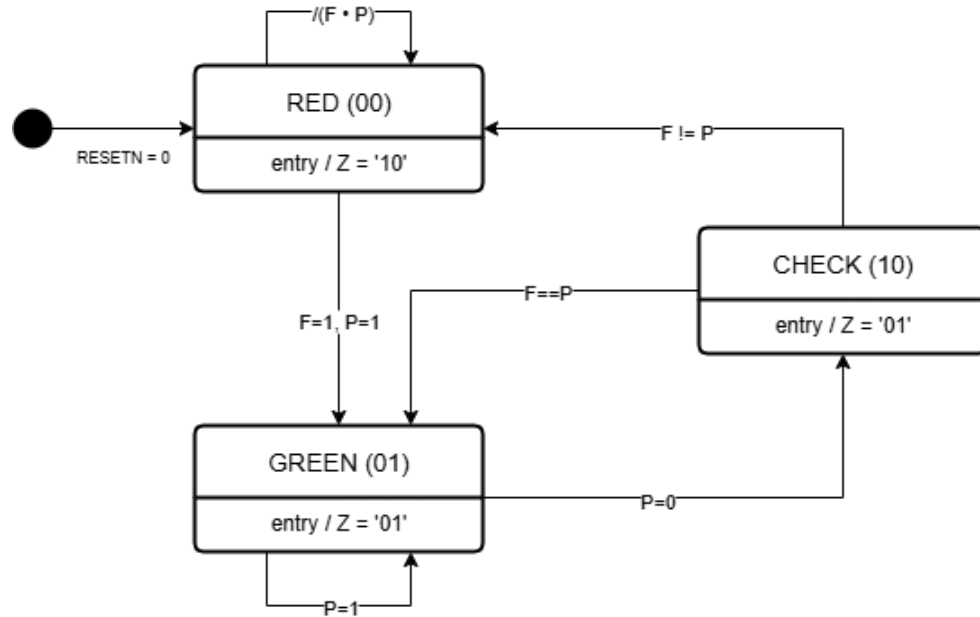


Figure 1. UML state diagram for a production line monitoring system, with the MSB (left bit) denoting RED (out of sync) output and LSB denoting GREEN (all-clear) output.

Current State	Q1	Q0	feeding	packing	Next State	Q1+	Q0+	grn	red
RED	0	0	0	0	RED	0	0	0	1
RED	0	0	0	1	RED	0	0	0	1
RED	0	0	1	0	RED	0	0	0	1
RED	0	0	1	1	GREEN	0	1	0	1
GREEN	0	1	0	0	CHECK	1	0	1	0
GREEN	0	1	0	1	GREEN	0	1	1	0
GREEN	0	1	1	0	CHECK	1	0	1	0
GREEN	0	1	1	1	GREEN	0	1	1	0
CHECK	1	0	0	0	GREEN	0	1	1	0
CHECK	1	0	0	1	RED	0	0	1	0
CHECK	1	0	1	0	RED	0	0	1	0
CHECK	1	0	1	1	GREEN	0	1	1	0
unused	1	1	0	0	x	x	x	x	x
unused	1	1	0	1	x	x	x	x	x
unused	1	1	1	0	x	x	x	x	x
unused	1	1	1	1	x	x	x	x	x

Figure 2. Transition table for the UML state diagram as in Figure 1.

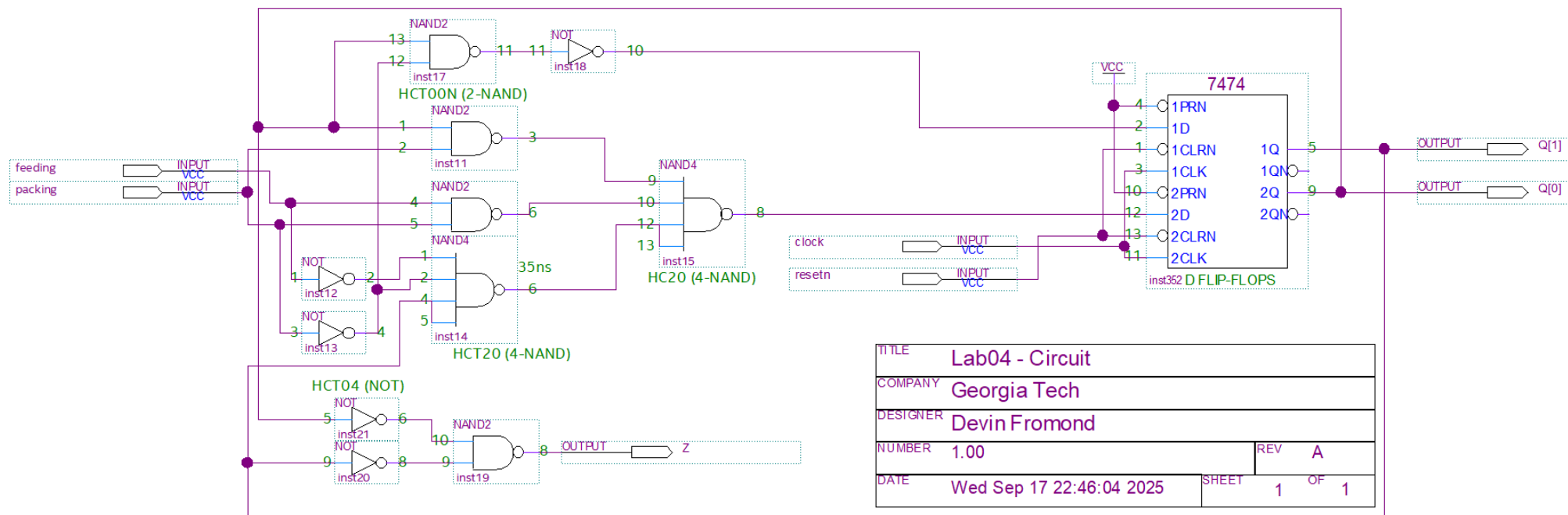


Figure 3. Schematic representing the state diagram as in Figure 1 with a 74HCT74 chip to hold sequential state logic.

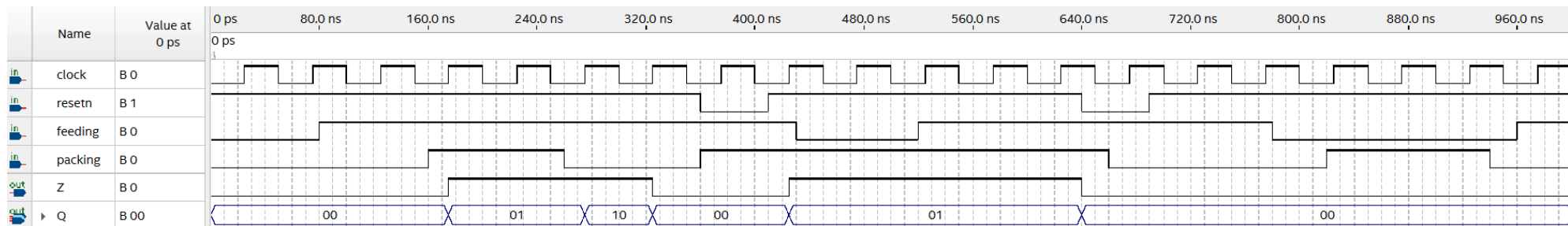


Figure 4. Simulation of the schematic as in Figure 3, highlighting the output (Z) and the state(s) (<Q1, Q0>).

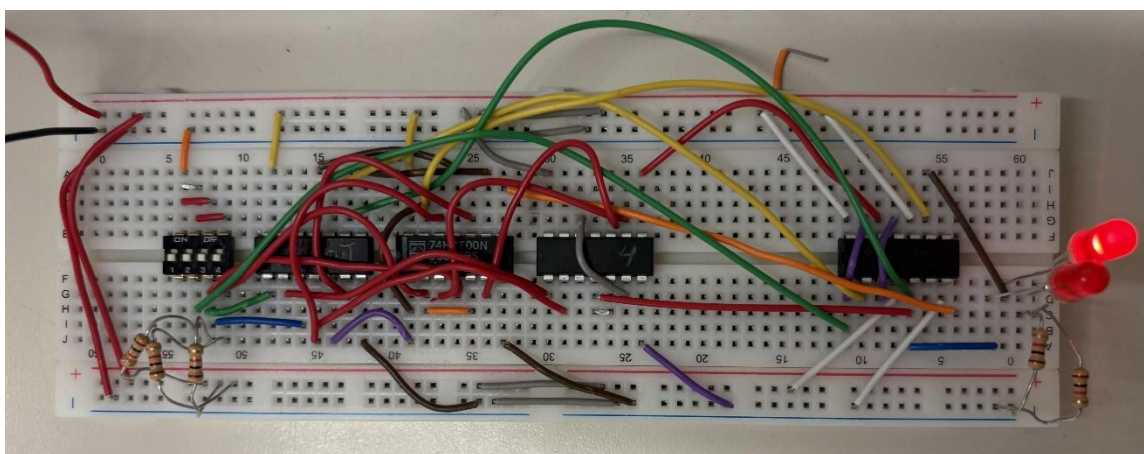


Figure 5. Breadboard prototype of Figure 3 schematic, with states shown via LEDs (top = MSB, bottom = LSB). Output omitted per lab instructions.

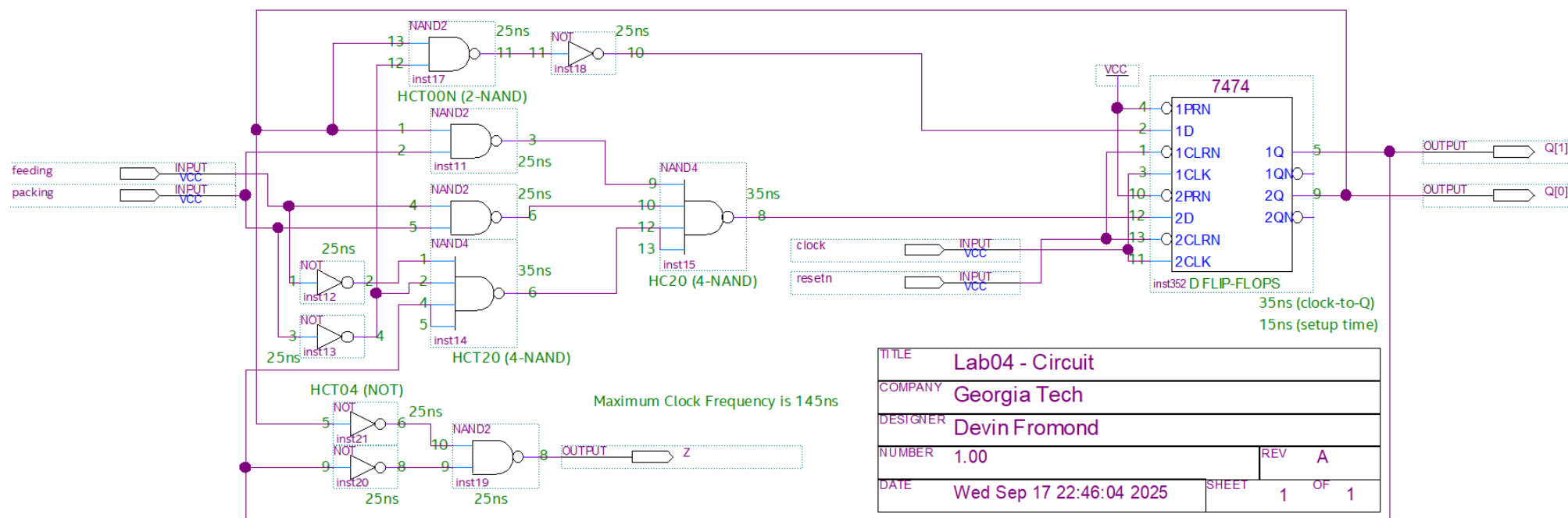


Figure 6. Schematic as in Figure 3 with worst-case propagation delays and maximum clock frequency calculated.